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Roll No

FT-228-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. State and prove the Pascal's law.
2. Find the displacement thickness δ^* , the momentum thickness θ and energy thickness δ^{**} for the velocity distribution in boundary layer given by $\frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2$.
3. A viscous flow is taking place in a pipe of diameter 100 mm. The maximum velocity is 2 m/s. Find the mean velocity and radius at which it occurs. Also calculate the velocity at 30 mm from the wall of the pipe.
4. a) Name the different forces present in a fluid flow. For the Euler's Equation of motion, which forces are taken into consideration.
b) Explain the classification of orifice and mouthpieces based on their shape, size and sharpness.
5. What is a venturimeter? Derive an expression for the discharge through a venturimeter.

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6. State Buckingham's π theorem. Why this theorem is considered superior over the Rayleigh's method analysis.

OR

Starting with Euler's equation of motion along a streamline and integrate it to obtain Bernoulli's equation. State all assumptions made.

7. What is Hagen Poiseuille's formula? Derive an expression for Hagen Poiseuille's formula.
8. Define : (Any two)
- a) Turbulent Boundary layer
 - b) Boundary layer thickness
 - c) Laminary sub layer
 - d) Laminar boundary layer
